

## Problems Related to Algebraic Number Theory

1. If  $k$  is a positive integer and  $p=4k+1$  is prime, prove that  $k^k \equiv 1 \pmod{p}$ .
2. You can write **ordered** pairs of integers on the blackboard according to this rule: Initially, there is only one pair  $(1, 2)$ . If  $(a, b)$  is already written, you can write  $(-b, a+b)$ . If  $(a, b)$  and  $(c, d)$  are already written, you can write  $(a+b, c+d)$ . Is it possible to write  $(2022, 2023)$ ?
3. In the Fibonacci sequence, find all prime numbers  $p$  such that neither  $F_{p+1}$  nor  $F_{p-1}$  is divisible by  $p$ .
4. This is a remnant of one of my other problems. There are two main ways to divide the edges of a  $K_7$  into 7 copies of  $K_3$ . They are obviously isomorphic, but are the theories behind them totally unrelated?
5. Consider 1000 cards, which are numbered 0 through 999, each having one gold side and one black side, lying on parallel on a long table. Initially all cards show their gold sides. Two players, standing by the same long side of the table, play a game with alternating moves. Each move consists of choosing an integer  $n$  that is at least 5 and the number  $n$  card is showing its gold side, and turning the cards numbered  $n, n-3, n-5$ , so those which showed gold now show black and vice versa. The last player who can make a legal move wins. (a) Does the game necessarily end? (b) Does there exist a winning strategy for the starting player?

6. Prove that the Ramsey numbers  $R(4,4)=18$  and  $R(3,3,3)=17$ .

7. Find all polynomials  $P(x)$  with real coefficients such that

①  $P(x)P(x+1) = P(x^2)$  ; ②  $P(x)P(x+1) = P(x^2 + 1)$  .

8. Prime numbers  $p$  and  $q$  are such that  $p$  is a primitive root mod  $q$ , and  $q$  is a primitive root mod  $p$ . Prove that in base  $p+q$ , one period of  $\frac{1}{p}$  and one period of  $\frac{1}{q}$  contain each digit from 1 to  $p+q-2$  exactly once.

9. Pick as many number as possible from 1 to 300 such that the sum of any four distinct numbers among them is not divisible by 11, and calculate the number of ways to pick the maximum amount of numbers.

10. There are  $n$  distinct integers, when you calculate their pairwise sums (for a total of  $\frac{n(n-1)}{2}$  sums; not including the sum of a number and itself), and take the remainder of each sum divided by 8, the remainders 0 through 7 appear an equal number of times. Find all possible values of  $n$ .

11. For positive integers  $a < b < c < d < e$ , denote by  $S_k$  the sum of their  $k$ -th powers. It turns out that  $S_k$  is a multiple of  $S_1$  if and only if  $k$  is not divisible by 5.

Find the three smallest values that  $S_1$  can take.

12. Let  $p$  be a prime number larger than 7. Positive integers  $1 \leq a < b < c \leq p-1$  satisfy

$$a + b + c \equiv a^3 + b^3 + c^3 \equiv a^5 + b^5 + c^5 \equiv \frac{p-1}{2} \pmod{p}.$$

Prove that  $\{2a^2 - 1 \pmod{p}, 4a^3 - 3a \pmod{p}\} = \{b, c\}$ .

13. Let  $S$  be the reduced residue system mod 2024. How many subsets of  $S$  have half the size of  $S$ , and the product of any two numbers (can be equal) from that subset, is not also in that subset?

14. Does there exist a positive integer  $n$  such that  $p = 10^n + 9$  is prime, and you can pick half of the numbers from 1 through  $p-1$  and put them in a row in some order, such that for any two adjacent numbers  $x$  and  $y$ , either  $x^2 + 4xy - y^2$  or  $x^2 - 4xy - y^2$  is divisible by  $p$ .

15. An  $m \times n$  table is glued into a cylinder. The side with length  $m$  is closed to a cycle but the side with length  $n$  remains open-ended. Proof that it is possible to fill real numbers (not all zero) into the table, such that the number in each cell is equal to the sum of all its neighboring cells (i.e. have a common side) if and only if there exists positive integers  $k$  and  $l$ , where  $k$  is not a multiple of  $n+1$ , satisfying  $\cos \frac{2l\pi}{m} + \cos \frac{k\pi}{n+1} = \frac{1}{2}$ . What will happen if the table is just a rectangle rather than a cylinder?